

Reusable CA1 population-update directions during boundary and reward changes

Convergent evidence from two longitudinal datasets

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Abstract

Hippocampal maps change when boundaries or goals change, but these population updates need not be constructed independently at every location. We tested whether the cell-resolved direction of one CA1 change predicts another in two public longitudinal calcium-imaging datasets. In an internal-wall arena (5,413 neurons, seven mice), a wall-minus-open contrast predicted later activity at the same seam (mean correlation advantage 0.184) and at a non-overlapping seam 25 cm away ($\bar{\rho} = 0.148$); both effects were positive in every animal. Prediction at the new seam remained after behavioral adjustment ($\bar{\rho} = 0.119$), but was lower than a query-matched same-seam benchmark ($\bar{\rho} = 0.230$). In a virtual-track reward task, two same-signed 120 cm goal relocations at different absolute positions produced more similar registered-cell update fields than either produced with an opposite-signed 240 cm return move (specificity $S = 0.0481$, eight of eight confirmatory mice, exact $p = 0.003906$). Deconvolved events, drift controls, trial and transition-label nulls, preprocessing variants, cell subsamples, and registration sensitivities preserved this result. Speed and licking showed stronger displacement-specific structure than CA1. A nonlinear out-of-fold trial model reduced CA1 specificity by about half, yet the residual remained positive in seven of eight mice ($S = 0.0238$, $p = 0.0117$; event residual $S = 0.0153$, $p = 0.0195$). Across two independent tasks, CA1 population changes therefore preserve cell-resolved direction across locations, while remaining spatially graded and partly shared with behavior. The results support structured reuse, not a coordinate-free or behavior-independent update operator.

Keywords: hippocampus; CA1; place cells; boundaries; reward; population code; generalization; remapping; cognitive map

1 Introduction

Place cells relate an animal’s position to a distributed hippocampal activity pattern, yet a cognitive map is not only a catalogue of positions. It must also change when

a wall appears, a route closes, or a goal moves. These changes can be described as vectors: for the same registered cells, which responses rise, which fall, and where along the represented space those changes occur. If two related events produce aligned vectors, the second update contains structure already present in the first. If they do not, each event is represented by an independent remapping.

Environmental boundaries and goals provide complementary tests of this idea. Place fields are anchored by geometry and often occur at characteristic distances and directions from walls (O’Keefe and Burgess, 1996; Hartley et al., 2000; Barry et al., 2006; Lever et al., 2009). A wall-related response may therefore recur when the wall relation is instantiated at another coordinate. Reward-related CA1 fields likewise generalize across reward locations (Sosa et al., 2025a), but shared coordinates do not imply that the full cell-by-position change evoked by one reward move reappears during a later move. Both questions require longitudinal cell identity and held-out comparisons; population-average firing or separately recorded cells are not enough.

We reanalysed two public longitudinal dorsal-CA1 datasets. The first followed seven mice through repeated layouts built from internal walls on a common lattice (Lee et al., 2025). We estimated a wall-minus-open contrast at one seam and tested later activity at the same seam or at a non-overlapping seam. The second followed reward switches on a circular virtual track (Sosa et al., 2025a). Within each visual environment, two 120 cm reward moves shared a signed displacement but began and ended at different absolute locations; a 240 cm return move provided a distinct comparison. We froze the reward endpoint after a separate development mouse and tested it in eight confirmatory animals.

The two analyses converge on the same organization. Wall-related changes predicted a different physical location but were more accurate at the original location. Reward-related changes with the same signed displacement were more aligned than the return update in every confirmatory mouse. Trial-level speed and licking accounted for substantial reward specificity, while a smaller

CA1 component remained after nonlinear out-of-fold conditioning. Thus related spatial events share a population-update direction, with limits set by physical location and behavior.

2 Results

2.1 Two tests of reusable population-change directions

The two datasets differ in task, geometry, and estimator, so their effect sizes were not pooled. They implement the same logical test (Fig. 1). First, a change vector is estimated while preserving registered-cell identity. Second, the same spatial relation is found at another absolute coordinate. Third, alignment with the later change is compared with a matched alternative. In the wall task the change is wall minus open and the alternative is an exact- location or spatial control. In the reward task the change is post minus pre; two same-signed 120 cm moves are compared with an opposite-signed 240 cm move.

2.2 A held-out test of cell-resolved wall contrasts

Each candidate wall location was represented as an oriented seam between two adjacent lattice tiles. For each registered cell, environment, and seam, we estimated activity in a narrow local strip and subtracted the corresponding estimate from a bracketing familiar-square session. This square residual reduced stable place preference while retaining the environment-specific local change.

Training geometries supplied two cell-wise profiles at each seam: a mean residual when the seam was blocked by a wall and a mean residual when it was open. Their difference was the estimated wall-related contrast. The neural outcome from the later target geometry was never used to form that source contrast. Prediction was scored by a rank correlation across held-out registered cells. We report animal-level means and sign counts rather than treating cells or queries as independent biological replicates.

2.3 A local wall-open contrast recurs across global geometries

For exact-location reuse, the broader wall context was matched between training and held-out comparisons. Across 252 held-out predictions, the wall-derived profile correlated more strongly with later wall responses than did the open-derived profile. The mean wall-minus-open correlation advantage was 0.184 and was positive in all seven animals (Fig. 2A). The sign was also positive in all animals when horizontal and vertical seams were evaluated separately.

This is more specific than a population-average increase near walls. The predictor is indexed by registered

cell identity: cells with positive or negative values in the estimated local contrast tended to retain those signs when the same seam appeared in another global geometry. Because the seam was unchanged, this result addresses recurrence at one location, not generalization across locations.

2.4 A one-tile counterfactual sharpens exact-location reuse

Two environments in the sequence, denoted u and o , differed only in whether one lattice tile was blocked. Their contrast therefore isolates one local wall state more cleanly than comparisons among complex layouts. We estimated the $u - o$ cell-wise contrast at that focal seam and tested it in held-out i , l , and t geometries that independently instantiated wall or open states there.

Across 46 held-out predictions, the estimated vector correlated with wall outcomes ($\bar{\rho} = 0.197$) but not open outcomes ($\bar{\rho} = -0.028$), giving an advantage of 0.225, positive in all seven animals (Fig. 2A). The raw-rate version gave a similar advantage (0.243). A model that adjusted local bin estimates for speed, heading, elapsed time, and occupancy retained the result, both with time included (0.225) and omitted (0.229).

The effect was more evident in the nearest independently eligible distance band, 2.5–7.5 cm from the seam (mean advantage 0.233, seven of seven animals), than in a 17.5–22.5 cm band (0.069, six of seven; Fig. 2D). These bands had separately defined supports, making the near-far comparison descriptive rather than paired.

The sequence was fixed: u always followed o . A generic same-seam, unchanged-state later-minus-earlier placebo was opposite in sign (mean -0.042 , zero of seven positive), whereas the actual $u - o$ contrast was positive (mean 0.195, seven of seven) and remained so after the nuisance adjustment (mean 0.212). This argues against a generic drift explanation, but it cannot remove an environment-specific order effect because wall state and order were never randomized.

2.5 A wall-related population contrast generalizes across locations

For each eligible target, a wall-minus-open vector was estimated at a distinct source seam with the same signed wall normal. The source and target seams were 25 cm apart, their local analysis bins did not overlap, and source training sessions were disjoint from the target session. The nearest source-target bins were 5 cm apart for tangential translations and 15 cm apart for normal-axis translations. The test used a separate multi-geometry estimator rather than the $u - o$ contrast.

The translated source vector predicted the later target-wall residual with mean correlation 0.148, positive in all seven animals (Fig. 2B). It also outperformed a source-open predictor by 0.152, and was more aligned with the

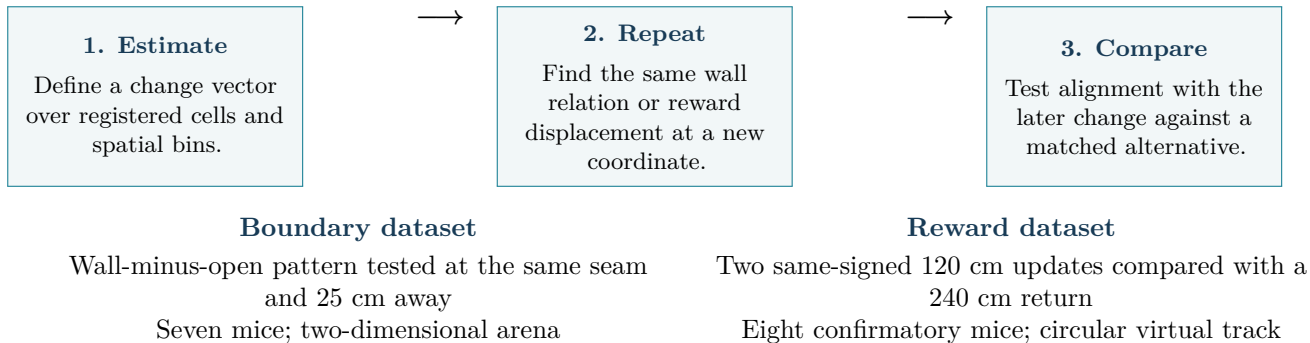


Figure 1: Common analysis logic across two independent datasets. Both analyses ask whether the direction of a registered-cell population change recurs when a spatial relation reappears at another absolute location. The estimators and numerical scales differ and are analysed separately.

target-wall response than with a training-derived target-open profile by 0.216, again positive in every animal. This cell-by-cell result is more specific than elevated activity at two wall locations: the signed contrast estimated at one coordinate predicted the response at another.

The exact-location benchmark was higher. On the same queries, cells, and support, it predicted the target response with mean correlation 0.230. Cross-location prediction was lower by 0.082, and only one of seven animals favored the shifted predictor. The response pattern generalizes, but remains strongly tied to its original coordinate.

2.6 Behavioral adjustment and spatial controls

Because animals may run differently near internal barriers, we re-estimated local neural response maps with covariates for speed, first and second circular harmonics of heading, elapsed time, and spatial sampling. The behavior-adjusted cross-location prediction remained positive in all seven animals ($\bar{\rho} = 0.119$). It correlated 0.172 more strongly with the target wall than with a training-derived target-open profile, and it exceeded a source-open predictor by 0.138; both differences were positive in all seven animals. Omitting elapsed time changed these values by less than 0.003.

An empirical spatial control replaced the correctly oriented source with source seams at the same 25 cm target-midpoint distance but with an incorrect signed wall normal. It retains the observed neural maps, registered-cell identities, target response, and source-target lag while breaking the correct wall relation. The correct-minus-incorrect advantage was 0.101, positive in all seven animals. Complete enumeration of the 2^7 animal-level sign assignments placed the observed mean in the upper 1/128 of that descriptive null. The result remained positive in every leave-one-animal-out summary.

This pooled comparison does not eliminate spatial continuity. When restricted to tangential translations with both a 25 cm midpoint distance and a 5 cm nearest-strip lag, the correct-orientation advantage was only 0.040,

positive in three of six animals (sign-flip tail fraction 0.25). Both correctly and incorrectly oriented tangential sources predicted the target. The pooled effect is therefore not an arbitrary equal-midpoint match. The most tightly distance-matched stratum, however, does not support a general orientation-specific rule.

Generic nearby place-field structure could still be shared by the source and target strips. A stricter within-session control was specified before its neural endpoint was inspected. For each wall target, a mirror-open seam in the same target session was matched for translation distance, axis, signed normal, relative bins, and registered cells. The geometry permits this match only for tangential translations. Although 104 labeled configurations were feasible, paired-support gates retained 72 triplets in five animals; two animals had no admissible control.

On this restricted subset, the globally demeaned source vector favored the wall target over the mirror-open target by 0.114, positive in all five eligible animals (Fig. 2C). Absolute source-to-wall correlation was positive in four of five animals ($\bar{\rho} = 0.068$), and a direct within-session wall-minus-open difference-vector score was positive in four of five ($\bar{\rho} = 0.077$). The raw-rate primary was positive in only three of five. The exact animal-level sign-flip tail fraction for the demeaned advantage was 1/32, and every leave-one-animal-out mean remained positive. In the supported demeaned tangential cases, the source contrast distinguished a wall target from an equally displaced open location. This control does not cover all seven animals, normal-axis translations, or raw-rate specificity.

A coherent source-cell identity permutation further showed descriptive upper-tail fractions below 0.01 in five animals and values of 0.069 and 0.060 in the other two. Registered-cell correspondence therefore contributes to the prediction, although the effect is not uniformly strong. Additional exploratory analyses of direction, composition, experience, and a former-barrier endpoint are reported in Supplementary Section 5 rather than treated as tests of the central phenomenon.

Across animal-level comparisons, cross-location prediction was consistently positive but lower than exact-

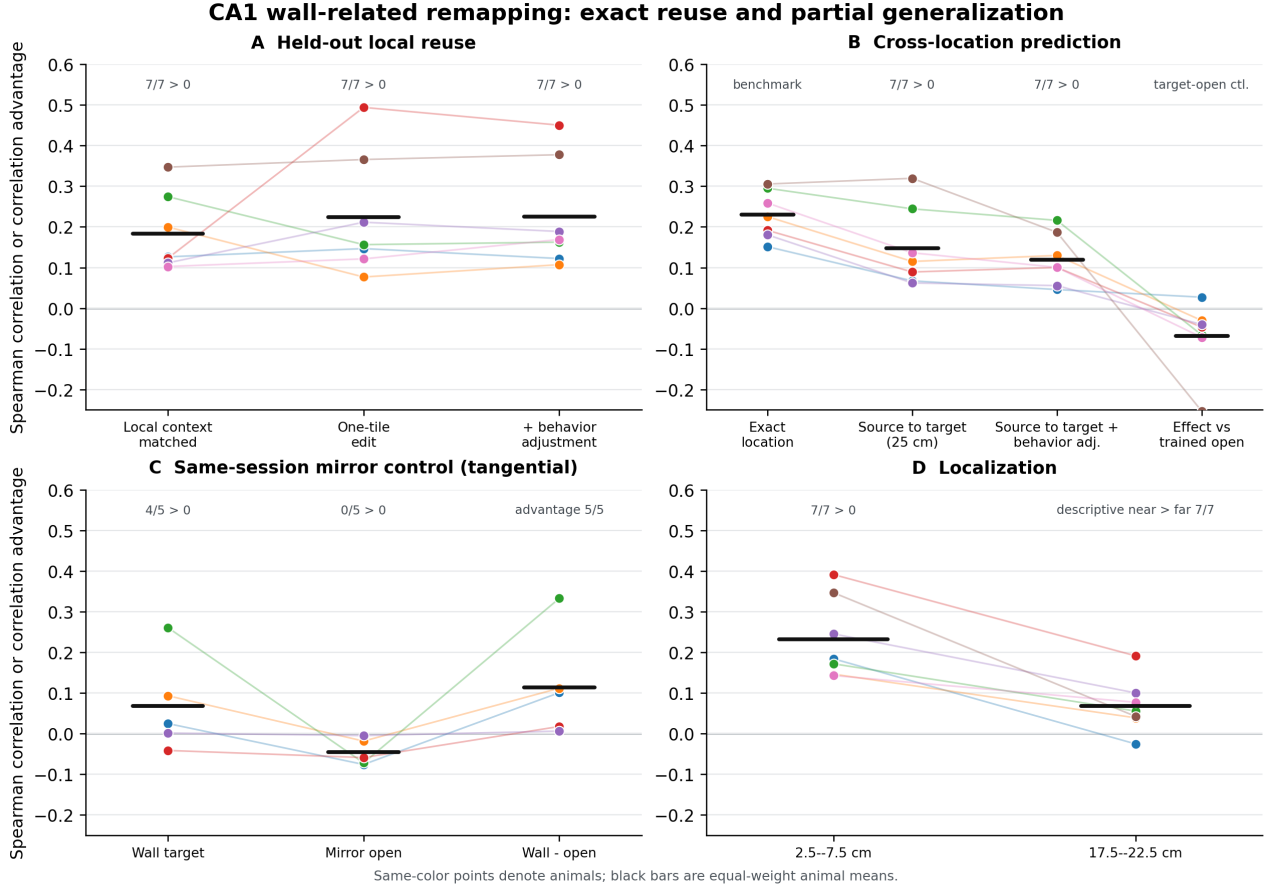


Figure 2: Exact reuse, cross-location prediction, and key controls. **A**, same-seam wall-minus-open prediction under three estimators. **B**, matched exact-location and source-to-target predictions, including behavioral adjustment and the target-open comparison. **C**, wall and mirror-open targets within the same session; five animals had paired support. **D**, descriptive near and far bands with separate eligibility. Points are animal means; bars are across-animal means.

location prediction. Spatial specificity was clear in the pooled orientation and mirror-open comparisons, but not in the strict tangential match (Fig. 3).

2.7 Repeated reward displacements reproduce a CA1 update direction

The reward dataset provides an independent test in a different geometry. Each confirmatory mouse contributed two three-day sequences, one in each visual environment. Within a sequence, two transitions moved the hidden reward 120 cm in the same signed direction, but from different starting positions. The third transition moved it 240 cm in the opposite direction. We aligned each post-minus-pre population update to the old reward and retained cells registered across all three days.

For each mouse and environment, we measured the Pearson correlation between the two short update fields and subtracted the mean of their correlations with the long return update:

$$S = r(\Delta_{120}^{(1)}, \Delta_{120}^{(2)}) - \frac{r(\Delta_{120}^{(1)}, \Delta_{240}) + r(\Delta_{240}, \Delta_{120}^{(2)})}{2}. \quad (1)$$

The mean same-displacement correlation was 0.120, compared with 0.071 and 0.073 for the two short-long pairs. Their difference was $S = 0.0481$ (mouse-bootstrap 95% interval [0.0234, 0.0757]), positive in all eight confirmatory mice (exact one-sided sign-flip $p = 0.003906$). Fourteen of the 16 mouse-environment values were positive. Deconvolved events gave $S = 0.0401$, again positive in eight of eight mice ($p = 0.003906$). Thus a same-signed reward move produced a reproducible update direction after the absolute start and end positions changed.

The comparison identifies the relation represented in the available design: the control move differs in both sign and distance. It therefore does not separate displacement direction from step length, and it tests the direction of the population update rather than whether its amplitude can be copied.

2.8 Reward-update specificity survives drift, null, and parameter tests

A pseudo-update constructed before each reward switch used the same mice, days, environments, registered cells, and transition schedule. Its mean specificity was 0.00235,

Partial generalization: the pattern crosses locations but remains place-bound

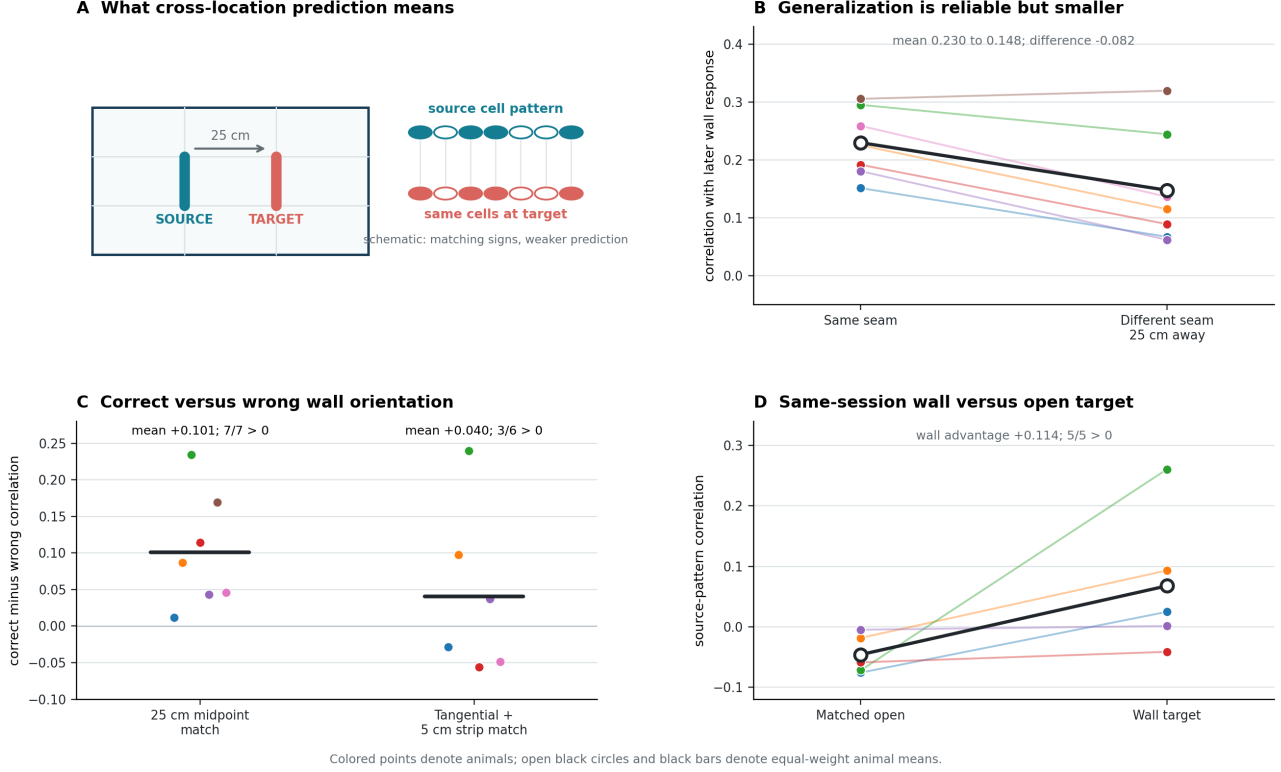


Figure 3: The central result and its spatial limits. **A**, schematic of a source wall-related cell pattern tested at a target seam 25 cm away. Filled and open circles denote schematic signed cell responses, not additional measurements. **B**, paired exact-location and cross-location correlations in seven animals. **C**, correct-minus-wrong orientation effects for the pooled 25 cm control and the strictly matched tangential subset. **D**, source-pattern correlations with same-session mirror-open and wall targets in five eligible animals. Quantitative panels show globally demeaned animal-level estimates.

with four of eight mouse means positive ($p = 0.359$); the corresponding post-switch drift estimate was -0.00114 ($p = 0.578$). The three fixed-condition mice lacked public cross-day ROI identities, so they could not support the same registered-cell endpoint. Within each of their 18 available sessions, however, the direction of early drift did not recur later: the mean quarter-to-quarter correlation across the three mice was -0.00352 (Fig. 4D).

Two label-preserving nulls also separated the reward result from chance. A trial-membership null had mean 0.00010 and 95% interval $[-0.01161, 0.01308]$; none of 199 frozen draws reached the observed value (plus-one $p = 0.005$). A 200,000-draw repeated-transition-label null had mean -0.000006 , 95% interval $[-0.02570, 0.02775]$, and $p = 0.000055$ (Fig. 4B).

All named normalization and spatial-resolution variants retained a positive cohort mean (range 0.0431 – 0.0679). All 500 deterministic half-cell subsamples were positive (mean 0.0488 , descriptive 95% interval $[0.0379, 0.0610]$), every leave-one-mouse-out mean was positive, and all 27 registration-threshold combinations retained the sign. These checks address drift, arbitrary labels, cell sampling, preprocessing, and registration thresholds without changing the frozen biological unit.

2.9 Reward-update geometry is shared with behavior

After post-switch speed and licking were translated into the neural old-reward frame, both behaviors carried strong displacement-specific structure. The speed-only endpoint was $S = 0.2261$ and the lick-only endpoint was $S = 0.3246$; both were positive in eight of eight mice ($p = 0.003906$). A spatial aggregate linear adjustment left a positive dF/F mean ($S = 0.0253$) but only five of eight mouse means were positive ($p = 0.0547$), below its frozen six-of-eight consistency requirement. The neural result therefore cannot be described as unrelated to how the animals ran and licked.

We then conditioned neural activity on behavior trial by trial. At each of 45 positions, a Gaussian radial-basis kernel used speed and $\log(1+\text{lick})$ in the adjacent and current bins, whole-trial behavior, and missingness indicators to predict all registered cells. Five-fold cross-fitting held out consecutive trial blocks within both pre- and post-switch conditions; all feature scaling, kernel bandwidths, cell means, and regression weights were estimated without the held-out trials. Reward condition and displacement were not model inputs. We recomputed Eq. 1 from the out-of-fold residual activity.

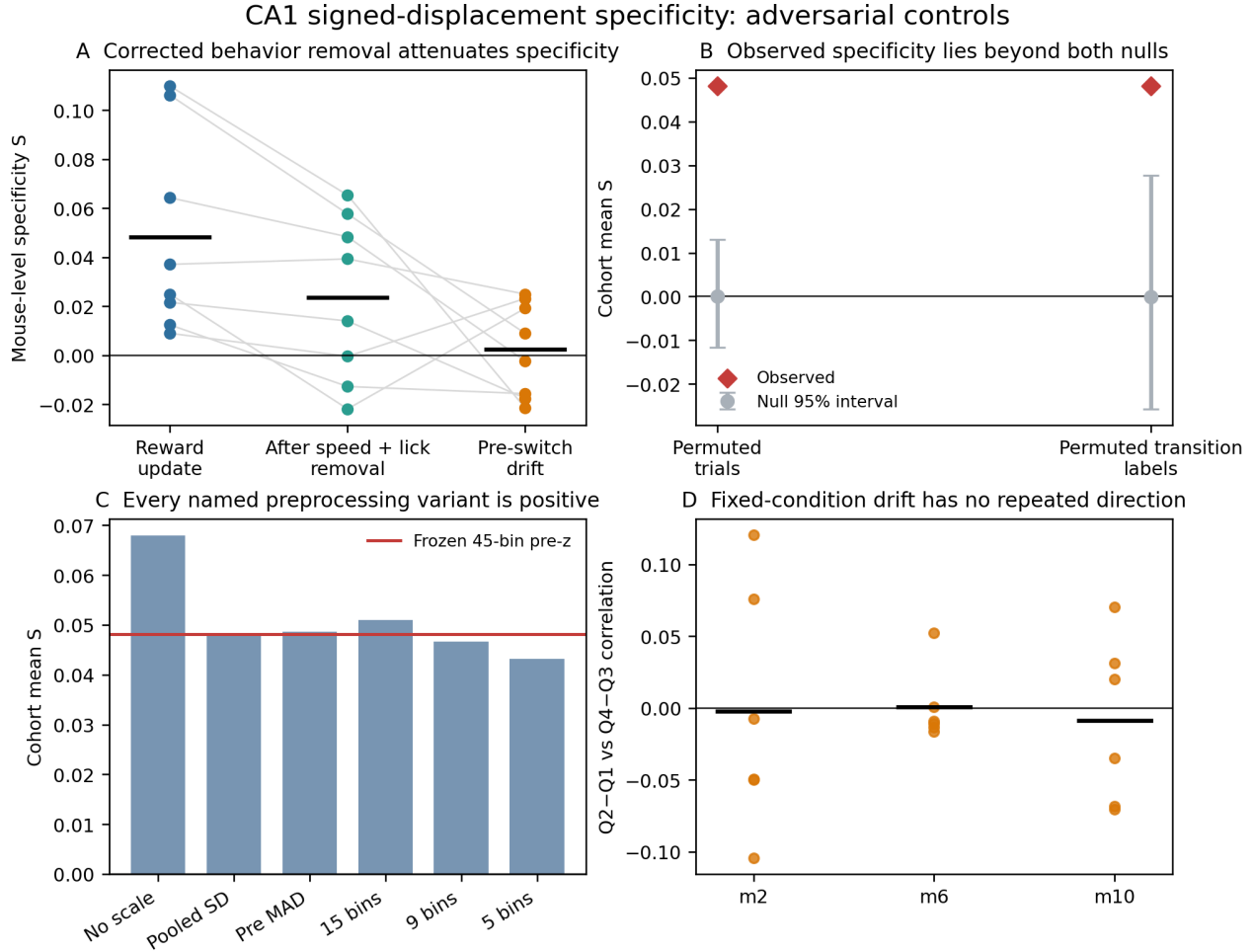


Figure 4: Controls for reward-displacement specificity. **A**, the original registered-cell endpoint, a corrected aggregate speed-and-lick residual, and pre-switch drift; points are mouse means. **B**, the observed cohort value and 95% intervals from trial-membership and transition-label nulls. **C**, named normalization and spatial-binning variants. **D**, within-session repeated-drift correlations in three fixed-condition mice. The aggregate behavioral residual in **A** did not meet its predeclared cross-mouse consistency gate; trial-level conditioning is shown in Fig. 5.

At the frozen primary regularization ($\lambda = 1$), residual dF/F specificity was $S = 0.0238$, positive in seven of eight mice ($p = 0.0117$; Fig. 5A). Residual events gave $S = 0.0153$, also positive in seven of eight mice ($p = 0.0195$). The dF/F mean was 49.5% of the original, and every mouse showed attenuation. Fixed $\lambda = 0.1$ and 10 sensitivities also retained positive dF/F means; the event result was positive at all three values and met the six-mouse consistency threshold at $\lambda = 0.1$.

Behavior explained a modest fraction of total held-out trial variance. At $\lambda = 1$, pooled mean-squared error improved by 1.03% for dF/F and 0.65% for events relative to a position-only cell mean. The more regularized $\lambda = 10$ model improved prediction by 2.77% and 2.56%, respectively, while removing less of the specificity signal. The flexible $\lambda = 0.1$ model overfit and performed worse than the position-only predictor, yet its dF/F residual specificity remained positive. The primary result therefore does not depend on selecting the model that yields the

most favorable neural endpoint.

2.10 Convergence across paradigms

The wall and reward analyses are not replications of an identical statistic. One predicts a local wall-related response across seams; the other compares full cell-by-position update fields across reward transitions. Their convergence lies in the registered-cell relation: a change direction estimated at one absolute location predicts a related change elsewhere, in seven of seven and eight of eight primary-animal summaries, respectively (Table 1). The differences are equally informative. Wall prediction declined across physical distance, and measured behavior accounted for about half of reward specificity. The shared phenomenon is structured reuse under a repeated spatial relation, not invariance to location or task.

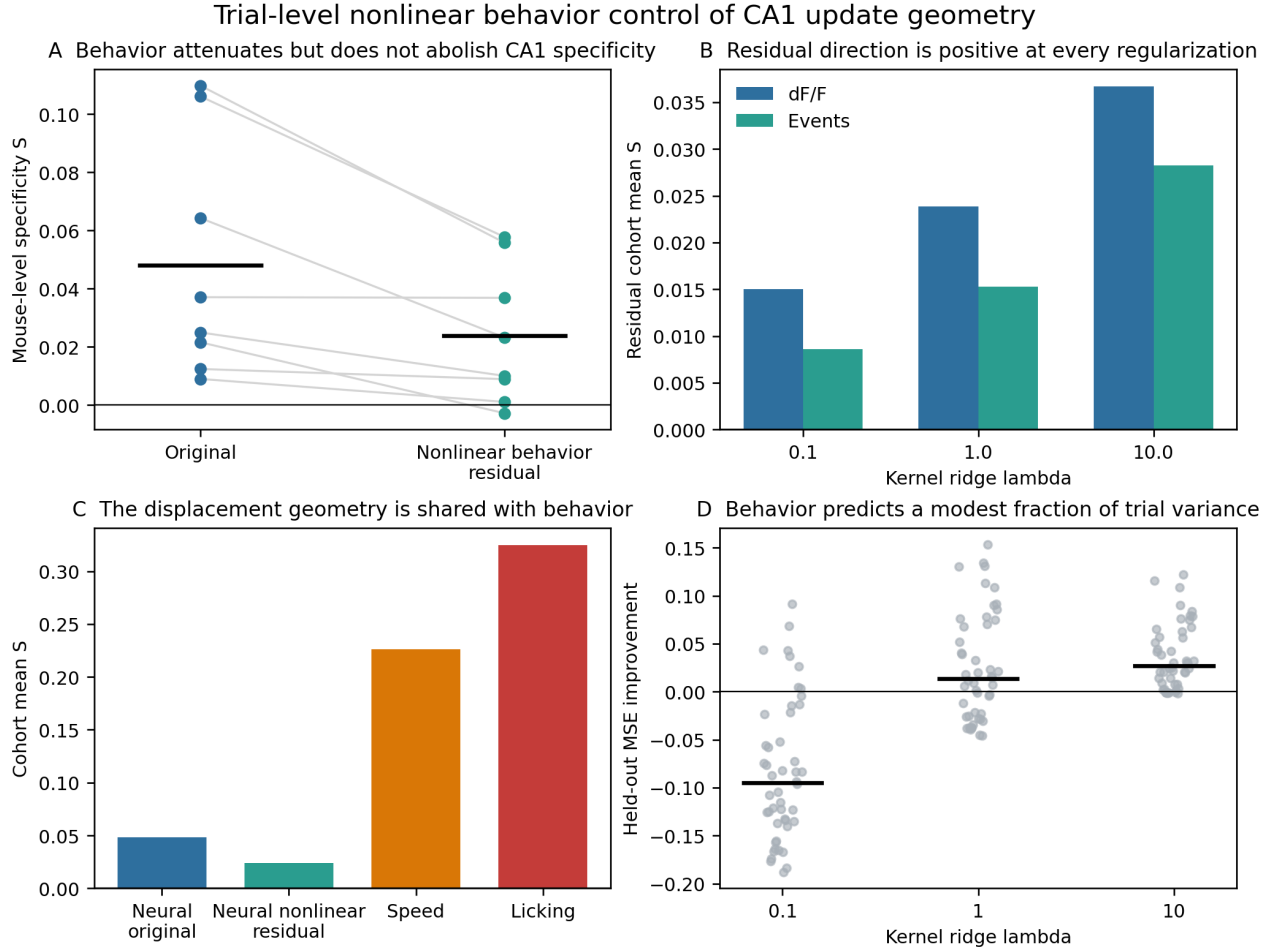


Figure 5: Trial-level relationship between behavior and CA1 reward updates. **A**, mouse-level specificity before and after nonlinear out-of-fold conditioning on speed and licking. **B**, residual dF/F and event specificity across the three fixed regularization values. **C**, the original neural endpoint, the primary nonlinear residual, and behavior-only endpoints. **D**, held-out neural prediction improvement for each session and regularization value; black bars show pooled values.

3 Discussion

Two independent longitudinal datasets show that CA1 population changes are related across absolute locations. In the boundary task, a cell-resolved wall contrast estimated at one seam predicted a later response 25 cm away. In the reward task, two same-signed relocations generated more aligned cell-by-position updates than a distinct return move. The effects were positive in every animal under each primary endpoint. This convergence extends the idea of a structured cognitive map from shared locations or average tuning to the direction in which a registered population changes.

The boundary result connects naturally to models in which place fields inherit distance- and direction-sensitive boundary inputs (Hartley et al., 2000; Barry et al., 2006). Barrier studies have shown local field formation, duplication, and geometry-dependent remapping (Rivard et al., 2004; Lee et al., 2025). Here the key observation is cell-resolved and predictive: the signed wall-minus-open pattern at a source coordinate anticipates which of the same

cells rise or fall at a later target coordinate. The target correlation of 0.148 was substantial but lower than the matched same-seam value of 0.230. The reusable component therefore coexists with a larger location-bound component.

The reward result adds a temporally separated form of reuse. Reward-related fields can generalize across locations (Sosa et al., 2025a); our analysis shows that the full update field produced by one relocation is preferentially aligned with the update from a second matching relocation. Its controls are important because any ordered sequence of recording days can produce apparent similarity. Registered pre-switch changes, fixed-session drift, permuted trials, and permuted transition labels did not reproduce the effect, and the result did not depend on one normalization, spatial resolution, cell sample, or registration threshold.

Behavior is part of the reward-related geometry. Speed and licking had much larger specificity values than the neural population, and nonlinear trial-level conditioning reduced neural specificity in every mouse. The remain-

Table 1: Evidence for reusable CA1 population-change directions. Values are means across animal-level estimates within each dataset. Numerical scales differ across estimators and are not pooled.

Dataset	Test	Mean	Sign	Supported interpretation
Wall	Context-matched exact seam	0.184	7/7	Local wall-open population contrast recurs across layouts.
Wall	Single-tile exact seam	0.225	7/7	Exact-location reuse remains in a cleaner local counterfactual.
Wall	Cross-location source effect	0.148	7/7	A cell-resolved wall contrast predicts a different location.
Wall	Query-matched exact effect	0.230	7/7	Same-location prediction is stronger.
Wall	Behavior-adjusted cross-location effect	0.119	7/7	The cross-location result remains after behavioral adjustment.
Wall	Mirror-open advantage	0.114	5/5	Demeaned tangential subset favors a wall over an equally displaced open seam.
Reward	Original dF/F specificity	0.0481	8/8	Same-signed 120 cm moves share more update direction than the return move.
Reward	Deconvolved-event specificity	0.0401	8/8	The result is preserved in an independently processed signal.
Reward	Nonlinear dF/F residual	0.0238	7/8	A smaller CA1 component remains after out-of-fold speed and lick conditioning.
Reward	Speed-only / lick-only	0.226/0.325	8/8	The displacement relation strongly structures behavior.

ing dF/F and event effects show that measured speed and licking do not exhaust the registered-cell relation. They do not establish a behavior-independent neural operation. Running and licking can be causes, consequences, or parallel readouts of a latent task-state change, and unmeasured variables such as pupil state, reward consumption, acceleration, and engagement remain possible.

The two datasets place complementary limits on the interpretation. Smooth or repeated place fields can correlate at nearby wall seams. Non-overlapping strips, square residuals, target-open specificity, an equal-midpoint orientation null, and a mirror-open target constrain this explanation, but the fully distance-matched tangential comparison was heterogeneous and the mirror control covered only five animals. The wall sequence was fixed rather than randomized. In the reward task, the available comparison confounds displacement sign with distance, and the fixed-condition animals lack public cross-day cell identities. Both reanalyses select cells stable enough to be registered longitudinally. Finally, the wall and reward statistics have different scales; their agreement is conceptual and cross-dataset, not a quantitative replication of one endpoint.

These observations define a middle level of organization. CA1 updates are not independent every time a related event occurs, yet the same vector cannot be applied unchanged anywhere. The data support reusable directions of population change whose expression depends on spatial position and behavioral state. They do not establish a symbolic map operation, portable update amplitude, composition, innateness, or a universal hippocampal primitive.

A direct prospective test could manipulate boundaries and goals independently within the same animals.

In a counterbalanced arena, a wall and a hidden goal would each move with repeated signed displacements at different absolute locations, while matched no-change, opposite-direction, and same-distance open conditions are interleaved. Longitudinal CA1 imaging plus trial-level movement, licking, eye, and posture measurements would reveal whether boundary and goal changes occupy separable, additive, or interacting directions in the same population geometry. The present estimates provide realistic targets for its power analysis without assuming that the two existing effect sizes are numerically interchangeable.

4 Materials and methods

4.1 Dataset description

4.1.1 Boundary-layout dataset

We reanalysed the public longitudinal dorsal-CA1 calcium-imaging dataset of Lee et al. (2025, 2024). It contains seven mice, 5,413 longitudinal cell identities, and 207 recording sessions. Each mouse foraged in the same familiar square outer arena and in nine internal-wall layouts defined on a common spatial lattice. The released data include position and sampling, spatial rate maps, geometry labels, and cell-registration identities across sessions. Six mice completed three exposure cycles (31 sessions each), and one completed two (21 sessions). Within every cycle, the nine wall layouts occurred in a fixed order between familiar-square sessions. Consequently, environment and sequence position were not independently varied. The original study describes surgery, imaging, behavioral procedures, preprocessing, and cell registration;

no new animal experiments were performed here.

4.1.2 Reward-relocation dataset

We independently reanalysed the public dorsal-CA1 two-photon dataset of Sosa et al. (2025a,c). Mice ran on a 450 cm circular virtual track with three possible unmarked reward zones and two visual environments. The processed release contains trial-by-position dF/F, deconvolved events, speed, licking, reward metadata, and published cell annotations. Imaging masks and ROI indices used to reproduce cross-day cell matching came from DANDI 001361 (Sosa et al., 2025b); behavioral variables came from the processed release because the audited NWB behavior arrays were not used.

Eleven mice performed reward switches and three remained in fixed conditions. Mouse 11 defined the endpoint in a development pilot and was excluded from confirmation. A metadata-only registration audit excluded mice 17 and 18 because curated NWB cell order could not be mapped exactly to the processed activity axes. The confirmatory cohort therefore comprised mice 3, 4, 7, 12, 13, 14, 15, and 19. Each provided two within-environment three-day sequences (days 3, 5, 7 and days 10, 12, 14). Day 8 was excluded because reward and visual environment changed together. Primary cell selection used only cross-day registration and training-only measurement reliability; published reward-relative labels did not select cells.

4.2 Spatial representation and oriented seams

The arena was expressed in a common allocentric coordinate system. A candidate internal-wall location was represented as an oriented seam $s = (p \rightarrow q)$ between adjacent lattice tiles, with a signed normal from tile p to tile q . The binary state $z_{e,s}$ indicated whether environment e blocked that adjacency. Local strips were partitioned into 5 cm spatial bins at defined distances from the seam. Analyses used only bins and registered cells satisfying the relevant occupancy, activity, and common-support gates. Source and target strips in the cross-location analysis shared no 5 cm bin.

4.3 Local response vectors

For registered cell i , environment e , and seam s , let $r_{i,e,s}$ be activity averaged over the eligible local bins. Let $b(e)$ denote the paired familiar-square session. We subtract that square baseline and collect the changes across the n shared cells:

$$\begin{aligned} \delta_{i,e,s} &= r_{i,e,s} - r_{i,b(e),s}, \\ \boldsymbol{\delta}_{e,s} &= (\delta_{1,e,s}, \dots, \delta_{n,e,s})^\top. \end{aligned} \quad (2)$$

For globally demeaned analyses, the vector mean was removed before correlation. The resulting score depends

on relative cell-specific changes rather than a population-wide rate shift.

4.4 Training profiles and held-out scoring

For a target query (e^*, s^*) , training environments were selected without using the neural response of e^* . At a source seam s , the wall and open profiles were

$$\begin{aligned} \boldsymbol{\mu}_{W,s} &= \text{mean}_{e \in \mathcal{T}: z_{e,s}=1} \boldsymbol{\delta}_{e,s}, \\ \boldsymbol{\mu}_{O,s} &= \text{mean}_{e \in \mathcal{T}: z_{e,s}=0} \boldsymbol{\delta}_{e,s}. \end{aligned}$$

computed on the common registered-cell support. The estimated source effect was $\mathbf{d}_s = \boldsymbol{\mu}_{W,s} - \boldsymbol{\mu}_{O,s}$. The held-out outcome was $\boldsymbol{\delta}_{e^*,s^*}$. Scores were Spearman correlations across registered cells. Wall-open advantages compare correlations within the same query and cell support.

The exact-location estimator set $s = s^*$. Its context-matched version restricted training comparisons so the broader internal-wall context was balanced between the wall and open profiles. The reported exact-location benchmark for cross-location prediction was recomputed on the exact queries, sessions, cells, and local supports eligible for the shifted analysis.

4.5 Single-tile counterfactual

The u and o geometries differed at one lattice tile: u blocked tiles $\{4, 5\}$, whereas o blocked only tile $\{4\}$. We formed the cell-wise $u - o$ contrast at the focal seam and evaluated wall and open outcomes at that seam in held-out i , l , and t geometries. Near (2.5–7.5 cm) and far (17.5–22.5 cm) bands were screened separately, so their difference is descriptive. For the order-placebo analysis, we formed later-minus-earlier contrasts at unchanged seams with the same processing and compared their signs with the actual $u - o$ contrast.

4.6 Cross-location prediction

For each eligible target seam, the source seam was a non-overlapping lattice seam 25 cm away with the same signed wall normal and admissible wall/open training support. Source training and target outcome sessions were disjoint. The primary shifted score was

$$\rho_{\text{shift}} = \rho_S(\mathbf{d}_{s_{\text{source}}}, \boldsymbol{\delta}_{e^*, s_{\text{target}}}).$$

Secondary specificity scores compared the source effect with a training-derived open profile at the target and compared source-wall with source-open predictors. The exact-location benchmark used $\mathbf{d}_{s_{\text{target}}}$ on the identical held-out query support. Stored-map and behavior-adjusted audits agreed on 405 target queries and 658 source pairs, including their session and seam identities and their counts of common cells and bins. That audit did not hash the complete cell-ID or bin-coordinate arrays.

4.7 Behavior-adjusted response estimates

We estimated activity jointly with spatial-bin indicators and nuisance covariates for instantaneous speed, sine and cosine of heading, second heading harmonics, and elapsed time. Spatial effects were evaluated at a common reference covariate distribution calibrated from the first familiar-square session. The no-time sensitivity omitted elapsed time but retained the other covariates. Target-session activity was used to estimate nuisance coefficients for that target map, but target neural outcomes never entered source-profile training.

4.8 Same-session mirror-open control

The mirror-open seam was specified before its neural endpoint was inspected. For each target wall, it lay in the same target session and matched the source-wall pair in translation distance, translation axis, signed normal, relative spatial bins, and registered cells. Both source-to-target nearest-bin distances were 5 cm. Only tangential translations admitted this exact geometric pairing. Primary scoring used globally demeaned source and target vectors. Eligibility required a complete source, wall-target, and open-target triplet on common support.

4.9 Spatial-null calibration

We calibrated two empirical spatial controls at the animal level. The first compared correctly oriented source seams with incorrectly oriented sources at the same 25 cm target-midpoint distance. A tangential stratum additionally matched translation axis and the 5 cm nearest-strip lag. The second compared the wall target with the same-session reflected-open target described above. Both controls retain the observed neural maps, including their place-field smoothness. For each paired animal-level contrast, we enumerated all 2^N sign assignments and calculated the fraction of null means at least as large as the observed mean. These are descriptive exact sign-flip calibrations, not confirmatory population p -values.

4.10 Cell-identity permutation

For each animal, we generated 999 coherent random mappings of source-cell identities while preserving query geometry, target vectors, support sizes, and the mapping across all queries in that permutation. The descriptive upper-tail fraction was $(1 + n_{\text{null} \geq \text{observed}})/(1000)$. Correlations and registration constraints limit exchangeability; the fractions were used as identity-specificity diagnostics rather than inferential population p -values.

4.11 Direction, composition, and experience analyses

Direction controls compared eligible source-target pairs with same or opposite signed wall normals while documenting translation axis, distance, overlap, and layout matching. No single comparison simultaneously matched all of these features across the full cohort.

For composition, we identified complete wall-state factorials in which two single-wall effects and their joint state could be estimated. The additive prediction was the sum of the two single effects. It was compared with an oracle best-single-component predictor using correlation and normalized prediction error. Only two repetitive factorials and five eligible animals were available.

Experience analyses contrasted transfer changes across exposure blocks using difference-in-differences variants. Because geometry, order, and exposure were not independently randomized, these estimates were treated as exploratory sensitivity analyses and not as evidence for learning.

4.12 Independent former-barrier analysis

We used the public neutral-barrier data from [Blair et al. \(2023b\)](#) to test whether the former barrier center retained an excess neural similarity relative to matched control locations after removal. The primary endpoint was a Pearson correlation among active cells, averaged as a former-barrier center-minus-control contrast. Its definition was written before the neural endpoint was run in the local project workflow. Spearman scoring and leave-one-animal-out summaries were sensitivities.

4.13 Reward-aligned population updates

For each reward-switch session, neural activity had dimensions trial by position by cell. We used 45 circular 10 cm bins and cells registered across all three days of a mouse-environment sequence. Each cell was centered and scaled using only its pre-switch samples, after which pre- and post-switch maps were averaged separately and circularly shifted so position zero marked the old reward. For transition k , the registered population update was

$$\Delta_k(x) = \bar{\mathbf{z}}_k^{\text{post}}(x) - \bar{\mathbf{z}}_k^{\text{pre}}(x).$$

Pearson correlations were calculated over common finite cell-by-position entries after flattening the field. The three-day specificity statistic was Eq. 1. It was calculated separately in the two visual environments and averaged within mouse. dF/F was primary; identically processed deconvolved events provided a signal-level robustness analysis.

The primary endpoint was redesigned after a declared mouse-11 development analysis rejected unit-amplitude prediction but identified the correlation-based contrast. The scoring code, eligibility rules, animal-level test, and

remaining confirmatory cohort were fixed before neural correlations from the eight confirmatory mice were computed. The mouse bootstrap resampled cells as blocks with all their position bins for a descriptive interval. Confirmatory inference used the eight mouse means in a complete one-sided 2^8 sign-flip test.

4.14 Reward-update drift and null controls

Registered pre-switch drift split each session’s pre trials into chronological halves, formed late-minus-early maps, and applied the same three-transition specificity statistic. Post-switch drift was calculated analogously. The processed release does not preserve cross-day identities for fixed-condition mice and DANDI 001361 contains no corresponding ROI assets. We therefore did not infer those identities from activity. The declared contingency split each fixed session into quarters and correlated Q2-minus-Q1 with Q4-minus-Q3; mouse-level means were reported descriptively.

The trial-membership null permuted trials into pseudo pre/post groups while preserving session structure and recomputed the full endpoint for 199 frozen draws. The transition-label null preserved the three observed update fields within each mouse-environment and randomly assigned which pair counted as the repeated transition; 200,000 draws were used. Monte Carlo tails used the plus-one correction. Predeclared sensitivities varied cell scaling, robust scale, and spatial resolution; resampled half the cells 500 times while keeping their spatial bins together; recomputed leave-one-mouse-out means; and applied 27 grids over minimum common-cell count, image correlation, and median ROI intersection-over-union.

4.15 Trial-level nonlinear behavior model

Upstream speed and lick profiles were centered separately on the old and new reward. We reordered their circular bins and translated every post-switch trial by the signed reward displacement (± 12 or ± 24 ten-centimeter bins), placing both neural and behavioral arrays in the old-reward frame. Behavior-only speed and lick updates then used the same three-transition specificity statistic as the neural maps.

The nonlinear model was fit separately at each position. Predictors were speed and $\log(1 + \text{lick rate})$ in the previous, current, and next circular bins; whole-trial means of both quantities; and a missingness indicator for each feature. A Gaussian radial-basis kernel ridge regression predicted all registered cells jointly. The bandwidth squared was the median positive pairwise squared distance among training feature vectors. Five consecutive folds were constructed within the pre and post conditions separately. Feature medians and scales, bandwidth, cell means, and regression weights were learned from four folds and applied to the held-out pre and post blocks. Missing fea-

tures were replaced by their training median and flagged. Reward condition, displacement, and trial number were excluded from the predictors.

Out-of-fold residuals retained trial, position, and registered-cell identity. We formed residual pre/post maps with the original pre-only neural scaling and recomputed Eq. 1. The frozen primary regularization was $\lambda = 1$; $\lambda = 0.1$ and 10 were fixed sensitivities. Predictive diagnostics compared held-out squared error with a position-specific training-cell mean that used no behavior. A separate aggregate linear control regressed each cell’s spatial update on corrected speed and lick updates. Its frozen cross-mouse consistency gate was evaluated independently of the nonlinear endpoint.

4.16 Statistical reporting and reproducibility

Animals were treated as the biological units. We report animal-level means, sign counts, query counts, and explicitly labeled descriptive permutation fractions. No cell-level or query-level observation was presented as an independent animal replicate. The wall analyses are exploratory: neither the original experiment nor the reanalysis was preregistered for the cross-location hypothesis. The reward endpoint is confirmatory relative to the declared local workflow: mouse 11 was used for development, and code plus gates were committed before scoring the remaining cohort. This process declaration is not a prospective registration of the original experiment.

All figure values derive from tracked machine-readable source files. The repository contains scripts, eligibility ledgers, result summaries, design locks, test suites, and an inspectable Git history. Accepted reward-control JSON files were reproduced byte for byte in independent reruns. The two datasets were analysed separately and no joint effect size or pooled animal test was computed.

5 Supplementary exploratory analyses

5.1 Direction and two-wall composition

Direction controls did not identify a general orientation rule. Same- versus opposite-signed normals for tangential translation favored the same direction in three of six eligible animals. A normal-axis comparison against opposite-facing or overlapping alternatives was positive in four of seven. A less controlled same- versus opposite-away comparison was positive in all seven, but it also changed strip distance and layout. These analyses support transfer between selected corresponding seams, not rotation- or direction-invariant transport.

Only two complete two-wall factorials were available, both involving the same bit and donut layouts. In five eligible animals, the demeaned additive predictor reached mean correlation 0.153, below an oracle best-

single-component predictor at 0.178. The additive advantage was -0.024 and positive in one of five animals. This sparse support provides no positive evidence for linear two-wall composition but cannot establish general non-compositionality.

5.2 Experience and former-barrier endpoints

A transfer-by-experience difference-in-differences estimate was positive under one stored-map specification (mean 0.045, four of four eligible animals), but raw-rate and stricter sensitivity analyses were unstable. Environment order, exposure number, and geometry were not independently varied, so the data do not identify experience-dependent stabilization or innateness.

In an independent neutral-barrier dataset (Blair et al., 2023b), the locked Pearson-active former-barrier center-minus-control endpoint was negative in all leave-one-animal-out summaries (animal mean -0.183 ; one of six animals positive). A Spearman sensitivity was near zero (mean 0.001; three of six positive). This endpoint showed no persistent local representational scar.

Data and code availability

The boundary-layout source data are available through the archive cited by Lee et al. (2024). Reward-task processed data are available through Figshare+ (Sosa et al., 2025c), with imaging and ROI assets in DANDI 001361 (Sosa et al., 2025b). The independent miniscope data used only for the supplementary former-barrier endpoint are available through Blair et al. (2023a). Boundary analysis code, derived source tables, audit reports, and manuscript sources are available at <https://github.com/asim-v/ca1-wall-update-transfer>. The reward-update analysis code, frozen protocols, machine-readable results, and audit logs are available at <https://github.com/asim-v/ca1-goal-update-transfer>. Neither repository redistributes the original raw recordings, which remain available from their source archives.

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References

- Caswell Barry, Colin Lever, Robin Hayman, Tom Hartley, Stephen Burton, John O’Keefe, Kate Jeffery, and Neil Burgess. The boundary vector cell model of place cell firing and spatial memory. *Reviews in the Neurosciences*, 17(1-2): 71–97, April 2006. doi: 10.1515/REVNEURO.2006.17.1-2.71.
- Garrett J. Blair, Changliang Guo, Shiyun Wang, Michael S. Fanselow, Peyman Golshani, Daniel Aharoni, and Hugh T. Blair. Data and analysis code for hippocampal place cell remapping during aversive learning. DANDI Archive, 2023a. URL <https://doi.org/10.5068/D12T2S>. Dataset and analysis code.
- Garrett J. Blair, Changliang Guo, Shiyun Wang, Michael S. Fanselow, Peyman Golshani, Daniel Aharoni, and Hugh T. Blair. Hippocampal place cell remapping occurs with memory storage of aversive experiences. *eLife*, 12:e80661, July 2023b. doi: 10.7554/eLife.80661.
- Tom Hartley, Neil Burgess, Colin Lever, Francesca Cacucci, and John O’Keefe. Modeling place fields in terms of the cortical inputs to the hippocampus. *Hippocampus*, 10(4): 369–379, 2000. doi: 10.1002/1098-1063(2000)10:4<369::AID-HIPO3>3.0.CO;2-0.
- J. Quinn Lee, Alexandra Keinath, and Mark P. Brandon. Mouse CA1 calcium imaging and behavioural dataset in 3x3 geometric morph paradigm. Zenodo, version v1, 2024. URL <https://doi.org/10.5281/zenodo.13993254>. Dataset.
- J. Quinn Lee, Alexandra T. Keinath, Erica Cianfarano, and Mark P. Brandon. Identifying representational structure in CA1 to benchmark theoretical models of cognitive mapping. *Neuron*, 113(2):307–320.e5, January 2025. doi: 10.1016/j.neuron.2024.10.027.
- Colin Lever, Stephen Burton, Ali Jeewajee, John O’Keefe, and Neil Burgess. Boundary vector cells in the subiculum of the hippocampal formation. *The Journal of Neuroscience*, 29(31):9771–9777, August 2009. doi: 10.1523/JNEUROSCI.11319-09.2009.
- John O’Keefe and Neil Burgess. Geometric determinants of the place fields of hippocampal neurons. *Nature*, 381(6581): 425–428, May 1996. doi: 10.1038/381425a0.
- Bruno Rivard, Yu Li, Pierre-Pascal Lenck-Santini, Bruno Poucet, and Robert U. Muller. Representation of objects in space by two classes of hippocampal pyramidal cells. *The Journal of General Physiology*, 124(1):9–25, July 2004. doi: 10.1085/jgp.200409015.
- Marielena Sosa, Mark H. Plitt, and Lisa M. Giocomo. A flexible hippocampal population code for experience relative to reward. *Nature Neuroscience*, 28:1497–1509, June 2025a. doi: 10.1038/s41593-025-01985-4.
- Marielena Sosa, Mark H. Plitt, and Lisa M. Giocomo. A flexible hippocampal population code for experience relative to reward. DANDI Archive, Dandiset 001361, 2025b. URL <https://doi.org/10.48324/dandi.001361/0.250609.2249>. NWB dataset, version 0.250609.2249.
- Marielena Sosa, Mark H. Plitt, and Lisa M. Giocomo. Data for: A flexible hippocampal population code for experience relative to reward. Figshare+, 2025c. URL <https://doi.org/10.25452/figshare.plus.27138633>. Processed dataset, version 1.